Rahul Anant Ramane

Senior Software Engineer | GenAI, LLM Engineering and Agentic AI

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LinkedIn GitHub Portfolio

PROFESSIONAL SUMMARY

Senior Software Engineer specializing in Generative AI, LLM-powered applications, and agentic automation using Python. Experienced in building Retrieval-Augmented Generation (RAG) systems, multi-agent workflows, and AI-driven testing solutions using LangChain, LangGraph, FastAPI, and Django. Delivered automation solutions across banking, insurance, pension, and fintech projects, with reported 70–87% output accuracy and 40–80% effort reductions across specific workflows. Skilled in prompt engineering, vector search, REST API development, and integrating LLMs into enterprise automation workflows.

TECHNICAL SKILLS

Generative AI and LLMs: Generative AI, LLM Engineering, Retrieval-Augmented Generation (RAG), Embeddings, Vector Search, FAISS, ChromaDB, Agentic AI, Multi-Agent Systems, Prompt Engineering, LLM Evaluation, MCP, NLP
AI Frameworks and Application Development: LangChain, LangGraph, Spec Driven Development (SDD), Django, Streamlit
LLM Providers and Developer Tools: OpenAI API, Claude, GitHub Copilot, GitHub Copilot Custom Agents, GitLab Duo, Mistral, GPT OSS, Continue
Languages, APIs and Cloud: Python, Java, SQL, REST APIs, Azure
Version Control and Collaboration: Git, GitHub
Testing and Automation: Selenium, Playwright, Cucumber, TestNG, Tosca, REST-Assured, Postman, JMeter

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Capgemini India Mar 2026 – Present

End-to-End STLC Automation Using GitHub Copilot Custom Agents (POC)

- Designing an agentic AI workflow with custom GitHub Copilot agents to automate the Software Testing Life Cycle (STLC), from Jira requirement analysis to test execution, reporting, and dashboard generation.
- Building a traceable pipeline that validates Jira requirements, generates test scenarios and manual/automation test cases, validates outputs, and converts approved automation cases into framework-aligned executable scripts.
- Supporting automated test execution and result reporting; using Git and GitHub for version control and change management across the POC workflow.

Test Case to Automation Script Generation

- Reduced manual scripting effort by 75%, with 87% script generation accuracy, by building custom GitHub Copilot agents that retrieve test cases from Azure DevOps and generate framework-aligned Java Selenium scripts and test data.

Java Selenium and API Automation Framework

- Cut per-test-case execution time by 40% by architecting an AI-compatible Selenium and REST-Assured framework in Java for a secure banking environment.
- Achieved 80–85% XPath generation accuracy by integrating LLM-powered XPath generation using the Continue plugin with GPT OSS and Mistral models.

Analyst, Capgemini India Sep 2024 – Feb 2026

Tosca Test Case Migration Agent

- Reduced manual documentation effort by 80%, with 97–100% conversion accuracy, by building an LLM pipeline with a Django backend and REST APIs that converts Tosca artifacts into structured manual test cases.

Tosca to Playwright Script Generation

- Reduced scripting time by 70%, with 85% generation accuracy, by building custom AI agents with Tosca MCP and GitHub Copilot that convert Tosca test cases into executable Playwright scripts.

JMeter Script Generation Platform

- Built a Django and Streamlit platform that generates standardized JMeter scripts from natural-language requirements using LLMs, reducing manual performance scripting effort across teams.

PROJECTS

MindVault – AI Document Intelligence and RAG Chat System

Python, FastAPI, LangChain, Next.js, React, FAISS, OpenAI, FlashRank GitHub

- Built a document-grounded conversational RAG application with Python, FastAPI, LangChain, FAISS, and OpenAI, providing source-cited answers and real-time Server-Sent Events (SSE) streaming.
- Implemented a retrieval pipeline using multi-query expansion, BM25 and vector search, Reciprocal Rank Fusion (RRF), cross-encoder reranking, and context compression; designed pluggable LLM and vector-store interfaces to swap providers without changing core application logic.

QAgent – Multi-Agent QA Automation Pipeline

Python, LangGraph, FastAPI, Next.js, Playwright, Vector Database, WebSockets Demo

- Built a LangGraph multi-agent QA workflow with Python and FastAPI to convert business requirements into executable Playwright tests, using an LLM fallback chain and WebSocket-based token streaming for orchestration and real-time progress updates.

EDUCATION

Bachelor of Technology, Electronics and Telecommunication Engineering 2024

Pimpri Chinchwad College of Engineering (PCCOE), Pune, Savitribai Phule Pune University 70.58%

CERTIFICATIONS

Microsoft Azure AI Fundamentals (AI-901); IBM Generative AI; IBM Prompt Engineering